Experiment 7 – Data Cleaning and Exploratory Data Analysis Using Pandas

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# Data Cleaning and Exploratory Data Analysis using Pandas
import pandas as pd

data = {
    "Name": ["Anitha", "Bala", "Chitra", "Dinesh", "Ezhil", "Fathima"],
    "Department": ["CSE", "CSE", "ECE", "CSE", "ECE", "CSE"],
    "Attendance": [92, 85, None, 88, 95, 90],
    "Mark": [86, 91, 78, None, 95, 82]
}

df = pd.DataFrame(data)

print("--- Original Dataset ---")
print(df)
print("\nMissing Values:")
print(df.isnull().sum())

# Missing value treatment
df["Attendance"] = df["Attendance"].fillna(df["Attendance"].mean())
df["Mark"] = df["Mark"].fillna(df["Mark"].mean())

# Remove duplicate records
df = df.drop_duplicates()

print("\n--- Cleaned Dataset ---")
print(df)
print("\n--- Descriptive Statistics ---")
print(df.describe())

print("\n--- Department-wise Average Marks ---")
print(df.groupby("Department")["Mark"].mean().round(2))

print("\n--- EDA Summary ---")
print("Highest Mark :", df["Mark"].max())
print("Lowest Mark :", df["Mark"].min())
print("Average Mark :", round(df["Mark"].mean(), 2))
print("Average Attendance :", round(df["Attendance"].mean(), 2))


# Additional CSV-based program from Experiment 7
# Create students1.csv separately with the following contents:
# Name,Department,Attendance,Mark
# Asha,CSE,90,88
# Bala,CSE,80,76
# Charan,ECE,,92
# Divya,ECE,95,
# Eshan,CSE,85,84

# In Google Colab, the CSV creation cell is:
# %%writefile students1.csv
# (Then enter the CSV rows shown above in that cell.)

# Load and analyze the CSV file
filename = "students1.csv"
df = pd.read_csv(filename)

print("--- Original Dataset ---")
print(df)
print("\nMissing Values:")
print(df.isnull().sum())

# Fill missing values with column means
df["Attendance"] = df["Attendance"].fillna(df["Attendance"].mean())
df["Mark"] = df["Mark"].fillna(df["Mark"].mean())

# Remove duplicate records
df = df.drop_duplicates()
```

```
print("\n--- Cleaned Dataset ---")
print(df)
print("\n--- Descriptive Statistics ---")
print(df.describe())

print("\n--- Department-wise Average Marks ---")
print(df.groupby("Department")["Mark"].mean().round(2))

print("\n--- EDA Summary ---")
print("Highest Mark :", df["Mark"].max())
print("Lowest Mark :", df["Mark"].min())
print("Average Mark :", round(df["Mark"].mean(), 2))
print("Average Attendance :", round(df["Attendance"].mean(), 2))
```